

QRM II Graded Assignment (4)

Period 1, 2026

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```
getwd()

## [1] "/Users/lizedenhertog/Documents/GitHub/QRM-Assignment-1"

#install.packages("tidyverse")
#install.packages("renv")
#install.packages("moments")
library(tidyverse)

## — Attaching core tidyverse packages — tidyverse 2.
0.0 —
## ✓ dplyr      1.2.1      ✓ readr      2.2.0
## ✓ forcats   1.0.1      ✓ stringr    1.6.0
## ✓ ggplot2    4.0.3      ✓ tibble     3.3.1
## ✓ lubridate  1.9.5      ✓ tidyr      1.3.2
## ✓ purrr     1.2.2
## — Conflicts — tidyverse_conflict
s() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()     masks stats::lag()
## ⓘ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors

library(renv)

##
## Attaching package: 'renv'
##
## The following object is masked from 'package:purrr':
##
##   modify
##
## The following objects are masked from 'package:stats':
##
##   embed, update
##
## The following objects are masked from 'package:utils':
##
##   history, upgrade
##
## The following objects are masked from 'package:base':
```

```
##  
##      autoload, load, remove, use  
  
library(moments)
```

Introduction

This assignment is to be completed in groups of 4-5. Further, all students in your group need to be assigned to the same R tutorial group (Friday's tutorial). You can sign yourself up for a group on Canvas. Please do so. You can use the Discussion Board in Canvas if you do not have a group yet or if your group is incomplete.

The assignment has 5 parts, and each part corresponds to the course material of that week (with the exclusion of week 6, for which there is no R programming material).

You are supposed to hand in these assignments on Canvas at the following dates:

- **Deadline 1** *Sunday September 27th, at 23:59pm*: you are supposed to hand in weeks 1, 2, and 3 of this assignment. This will determine 18% of your overall course grade
- **Deadline 2** *Sunday October 11th, at 23:59pm*: you are supposed to hand in weeks 4, and 5 of this assignment. This will determine 12% of your overall course grade

The R tutorials (each Friday) will consist of two halves. During the first half, you will discuss the tutorial exercises. These can be downloaded separately from Canvas. During the second half, you can work on this graded assignment within your own group. The purpose is that you find out how to work with R for doing statistical analyses by yourself. The tutorial exercises are meant to teach you basic commands to get you started, but to answer the problem sets in this assignment, you might need to research your own solutions, and use functions and commands not described in the tutorial exercises. Learning how to solve your own research problems is integral part of learning R. When you and your group get stuck on how to approach an exercise, the hierarchy in finding your way is as follows:

- use the concepts from the tutorial exercises;
- use the cheat sheets available on Canvas;
- use Google, YouTube, StackOverflow, or another website;
- ask the teacher.

The use of generative AI is permitted and may result in a grade of 0. See the AI protocol in the course manual for details.

To answer the assignment, you can simply fill out this R markdown document. There are designated places which you can fill with R code. There are also designated spaces for you to answer each question. Often, the structure of an answer will be as follows. First, you type the R code in the designated box. This will show how you analyzed the data to get the

answer to the question. Below the box for the R code, you will then summarize your answer to the question, i.e. what are the conclusions that you draw from the data analysis?

When handing in, you are supposed to submit this .Rmd file, and a knitted version of this document. You can knit this document to pdf, word, or html. Knitting to pdf requires you to have a .tex distribution installed on your computer. Knitting to Word requires you to have Word installed.

The exercises are designed such that you should be able to finish the majority of them during the tutorial each week. If you are not able to finish them fully during that time, you are expected to work on it in your own time using the computers on campus or your own device. It is best to meet as a group in-person when working together. If you want to work remotely, github is a good platform to guarantee smooth collaboration. Alternatively, you can email this .Rmd file back and forth to one another as a group, but this is not recommended as it is more cumbersome.

We encourage you to keep your code blocks, printing statements, and final answers, as short as possible. In any case, there is a page limit of 6 pages per week, which encompasses the total length of this document which consists of the questions, your coding lines, and your answers. When your answers to questions of the respective week exceed this page limit, they will not be graded, resulting in zero points.

Each week consists of 1, 2, or 3 subquestions. The total amount of points you can earn per week is 20 points.

Week 1

1. Find the dataset “movies4.tsv” on Canvas. Describe your data set: How many observations does it have. How many variables are there? How many subjects? What consists of a subject?

```
movies <- read_tsv("movies4.tsv")

## Rows: 808 Columns: 19
## — Column specification —————
## Delimiter: "\t"
## chr  (8): keywords, original_language, title, genre, first_actor, first_a
ct...
## dbl  (10): index, budget, popularity, revenue, runtime, vote_average, vote
_c...
## date  (1): release_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this m
essage.

summary(movies)
```

```
##      index      budget      keywords      original_language
## Min.   :    1  Min.    :      0  Length   :808  Length   :808
## 1st Qu.:1160  1st Qu.:     22  N.unique :702  N.unique : 18
## Median :2342  Median : 15000000  N.blank  :  0  N.blank  :  0
## Mean   :2353  Mean    : 29842216  Min.nchar:  4  Min.nchar:  2
## 3rd Qu.:3510  3rd Qu.: 40000000  Max.nchar:106  Max.nchar:  2
## Max.   :4801  Max.    :30000000  NAs      : 87
##      title      popularity      release_date      revenue

## Length   :808  Min.    :2.388e-03  Min.    :1990-03-23  Min.    :0.000e+0
## N.unique :808  1st Qu.:4.898e+00  1st Qu.:2001-11-09  1st Qu.:0.000e+0
## N.blank  :  0  Median :1.270e+01  Median :2007-03-29  Median :1.456e+0
## Min.nchar:  2  Mean    :2.032e+01  Mean    :2006-05-22  Mean    :8.273e+0
## Max.nchar: 59  3rd Qu.:2.656e+01  3rd Qu.:2011-10-04  3rd Qu.:8.521e+0
## Max.    :2.020e+02  Max.    :2016-07-28  Max.    :1.520e+0
## runtime      vote_average      vote_count      genre
## Min.   :  0.0  Min.    : 0.000  Min.    :  0.00  Length   :808
## 1st Qu.: 93.0  1st Qu.: 5.500  1st Qu.:  54.75  N.unique : 13
## Median :103.0  Median : 6.200  Median : 240.00  N.blank  :  0
## Mean   :105.4  Mean    : 6.039  Mean    : 688.58  Min.nchar:  5
## 3rd Qu.:116.0  3rd Qu.: 6.800  3rd Qu.: 776.50  Max.nchar:11
## Max.   :192.0  Max.    :10.000  Max.    :12002.00  NAs      :  3
## release_year release_month      release_day      first_actor
## Min.   :1990  Min.    : 1.000  Min.    :  1.00  Length   :808
## 1st Qu.:2001  1st Qu.: 4.000  1st Qu.:  8.00  N.unique :555
## Median :2007  Median : 7.000  Median :15.00  N.blank  :  0
## Mean   :2006  Mean    : 6.708  Mean    :15.33  Min.nchar:  5
## 3rd Qu.:2011  3rd Qu.: 9.000  3rd Qu.:23.00  Max.nchar:23
## Max.   :2016  Max.    :12.000  Max.    :31.00  NAs      :  6
## first_actor_gender director_first_name director_gender
## Length   :808  Length   :808  Length   :808
## N.unique :  2  N.unique :357  N.unique :  2
## N.blank  :  0  N.blank  :  0  N.blank  :  0
## Min.nchar:  4  Min.nchar:  2  Min.nchar:  4
## Max.nchar:  6  Max.nchar:18  Max.nchar:  6
## NAs      :34  NAs      :  4  NAs      :43
```

```
n_distinct(movies$index)
```

```
## [1] 808
```

```
#Answer to question 1
```

The dataset “movies4.tsv”, renamed to movies in this project, contains 808 observations of 18 variables (excluding the index). A subject is a single, unique object in a dataset and the measured values associated with it. There are 808 subjects, which is equal to the amount of observations. This is because each observation has a unique index number and a unique title as well.

2. Which of the following types of variables are present in your data set? (i) nominal; (ii) ordinal; (iii); interval; (iv) ratio. If present, name one example of such a variable present in your data set.

```
str(movies)

## spc_tbl_ [808 × 19] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ index          : num [1:808] 2256 2473 1514 1512 395 ...
## $ budget         : num [1:808] 0.0 1.5e+07 3.2e+07 3.2e+07 8.5e+07 0.
## 0 6.0e+04 0.0 0.0 1.4e+07 ...
## $ keywords       : chr [1:808] NA "barcelona spain menage a trois aut
## hor" "gunslinger revenge prairie shootout pistol" "robbery double life dual i
## dentity small town indiana" ...
## $ original_language : chr [1:808] "en" "en" "en" "en" ...
## $ title          : chr [1:808] "The Oogieloves in the Big Balloon Adv
## enture" "Vicky Cristina Barcelona" "The Quick and the Dead" "A History of Vio
## lence" ...
## $ popularity      : num [1:808] 0.286 32.758 16.473 34.629 44.967 ...
## $ release_date    : Date[1:808], format: "2012-08-29" "2008-08-15" ...
## $ revenue         : num [1:808] 0.00 9.64e+07 1.86e+07 6.07e+07 1.94e+
## 08 ...
## $ runtime         : num [1:808] 88 96 107 96 136 92 93 97 138 149 ...
## $ vote_average    : num [1:808] 2.2 6.7 6.3 6.9 6.7 7.1 5.1 5.6 6.7 6.
## 1 ...
## $ vote_count      : num [1:808] 8 1020 427 832 1225 ...
## $ genre           : chr [1:808] "Family" "Drama" "Action" "Drama" ...
## $ release_year    : num [1:808] 2012 2008 1995 2005 2006 ...
## $ release_month    : num [1:808] 8 8 2 9 12 8 10 9 9 9 ...
## $ release_day     : num [1:808] 29 15 9 23 8 4 13 23 3 30 ...
## $ first_actor     : chr [1:808] "Christopher Lloyd" "Scarlett Johansso
## n" "Sharon Stone" "Viggo Mortensen" ...
## $ first_actor_gender : chr [1:808] "male" "female" "female" "male" ...
## $ director_first_name: chr [1:808] "Matthew" "Woody" "Sam" "David" ...
## $ director_gender  : chr [1:808] "male" "male" "male" "male" ...
## - attr(*, "spec")=
## .. cols(
## ..   index = col_double(),
## ..   budget = col_double(),
## ..   keywords = col_character(),
## ..   original_language = col_character(),
## ..   title = col_character(),
## ..   popularity = col_double(),
## ..   release_date = col_date(format = ""),
## ..   revenue = col_double(),
```

```
## .. runtime = col_double(),
## .. vote_average = col_double(),
## .. vote_count = col_double(),
## .. genre = col_character(),
## .. release_year = col_double(),
## .. release_month = col_double(),
## .. release_day = col_double(),
## .. first_actor = col_character(),
## .. first_actor_gender = col_character(),
## .. director_first_name = col_character(),
## .. director_gender = col_character()
## .. )
## - attr(*, "problems")=<pointer: 0xb48d4c2c0>
```

In this dataset, there are nominal, interval, and ratio variables present. An example of a nominal variable is the gender of the director, expressed as male, female, or NA. There is no order to these characteristic variables, making them nominal. An example of an interval variable is the release date. The difference between these dates is meaningful, but there is no true zero in dates. An example of a ratio variable is run time, the difference between run times is meaningful and zero minutes of run time would in fact mean the absence of run time.

3. A movie studio wants to know which types of movies are best at generating profit. The measure they use for this is the profitability ratio, which is defined as

$$PR = \frac{Revenue}{Budget}.$$

Perform the following steps to provide the movie studio with an analysis which corresponds to their request:

- a. Create the variable PR as the revenue of a movie divided by its budget. Report its mean, median, maximum, and minimum.
- b. As you can probably see from your answer in a.), some movies have a PR of “Inf”. What does this mean? Why are some movies assigned this value?
- c. As you can probably see from your answer in a.), some movies have a PR of “NA”. What does this mean? Why are some movies assigned this value?
- d. For the movies that have a PR of “Inf”, set its PR to “NA”. After doing this, report the mean, median, maximum, and minimum of PR. How do you interpret the mean?
- e. What does a PR of at least one mean? What is the fraction of movies that have a PR of at least one?
- f. Create a boxplot of the variable PR. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. Describe the shape of the boxplot. Give Q_1 , Q_2 , Q_3 and Q_4 . What does this tell you about the nature of making money in the movies industry?

For each step, you should provide first all the code you used to answer the question and then formulate an answer using full sentences.

Step a

```
#Creating the new variable PR
movies$PR <- movies$revenue / movies$budget

#Computing the summary statistics
mean <- mean(movies$PR, na.rm = TRUE)
median <- median(movies$PR, na.rm = TRUE)
max <- max(movies$PR, na.rm = TRUE)
min <- min(movies$PR, na.rm = TRUE)
```

For the new variable called PR, the mean is given as Inf, the median is approximately 1.70496, the maximum is also given as Inf, and the minimum is zero.

Step b

Some movies have a budget of 0. When you divide the revenue which greater than 0 by the budget of 0 it wil result in Inf. This means that the value is infinite, or at least too large for the computer.

Step c

NA means not available. It indicates that the value for the PR is missing. In the column PR a specific type of NA is present, which is the NaN (not a number). This is the result of dividing zero by zero, which is not possible and therefore not a number. This can be interpreted as an NA, because a zero in budget and revenue most likely is not the absence of a budget and/or revenue, but simply the unavailability of that value.

Step d

```
#Turning Inf into NA
movies$PR[is.infinite(movies$PR)]<- NA
#Computing the new summary statistics
new_mean <- mean(movies$PR, na.rm = TRUE)
new_median <- median(movies$PR, na.rm = TRUE)
new_max <- max(movies$PR, na.rm = TRUE)
new_min <- min(movies$PR, na.rm = TRUE)
```

Your Answer:

Now that the infinite values have been replaced with NA in the column PR, the mean is approximately 2.8646, the median is approximately 1.5358, the maximum is approximately 73.6715, and the minimum has remained 0. The mean can be interpreted as the fact that on average, a movie earned 2.6846 times the budget in revenue.

Step e

```
mean(movies$PR>=1 & !is.na(movies$PR))

## [1] 0.4566832
```

```
mean(movies$PR>=1, na.rm = TRUE)
```

```
## [1] 0.6059113
```

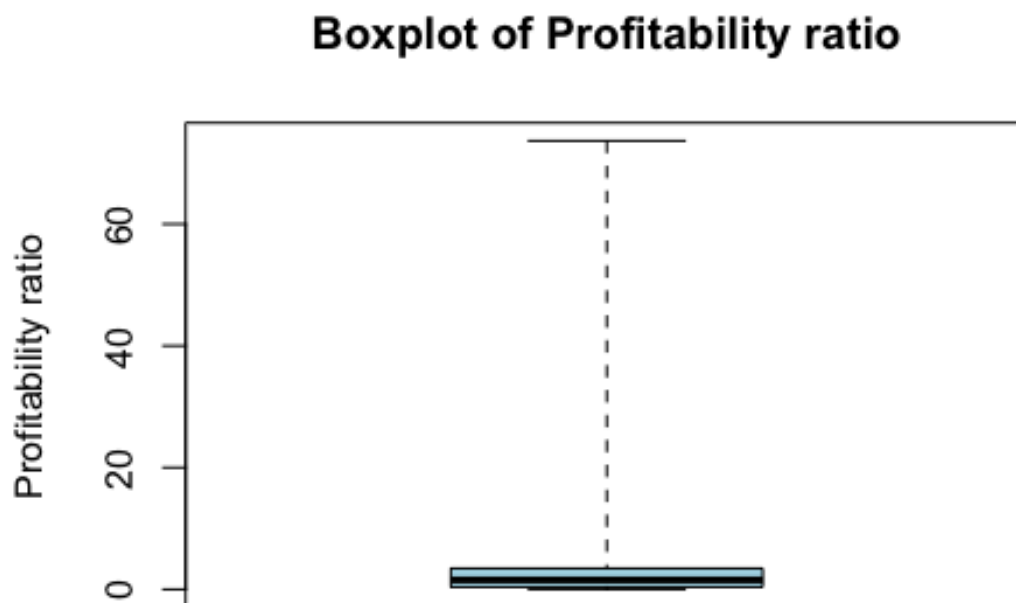
Your Answer:

A movie with a PR of at least one means that the movie is at least break even or has made a profit. A PR of one is the border between a movie that made a loss and a movie that made a profit. The fraction of movies with a PR of at least one is approximately 0.4567 when including the NAs as observations that do not fulfill the condition. When completely excluding the NAs in computing the fraction, the fraction becomes approximately 0.6059

Step f

```
#Computing the box plot
```

```
boxplot(movies$PR, main = "Boxplot of Profitability ratio", ylab = "Profitability ratio", col = "lightblue", range = 0, na.rm = TRUE)
```



```
#Computing the Quartiles
```

```
quantile(movies$PR, na.rm = TRUE)
```

```
##           0%           25%           50%           75%          100%  
## 0.0000000  0.3667304  1.5358255  3.4484473 73.6714634
```

Your Answer:

The boxplot is extremely unfairly distributed. The first 25% of the observations is barely visible and the interquartile range (25-75%) is very small as well. The highest 25% however, is extremely widespread. This illustrates that the profitability ratios are unfairly distributed between movies. The first quartile is 0.3667, the second quartile, also known as the median, is 1.5358, the third quartile is 3.4484, and the fourth quartile, also known as the maximum, is 73.6715 (all these values have been rounded down to four decimals). These numbers show that very few make a lot of money in the movie industry, and that most make significantly smaller profits. So, the nature the movie industry is that most barely make a profit or even make a loss, while only a few make huge profits.

Week 2

1 Is your dataset movies4.tsv the full population, or is it a sample of a larger population? If the latter, how would you describe the full population?

Your Answer

The dataset is a sample of the population. The dataset includes a limited amount of movies from 1990 until 2016. As it only includes 808 movies, which is a limited amount, as there were a lot more movies that were published in this timespan. The full dataset could be all the movies that have information on the variables included in the dataset (such as, runtime, director, etc.) published from 1990 until 2016.

2 Let M be the event that a movie director is male. Let R be the runtime of the movie.

- Using your data, calculate $P(M)$. What is $P(M')$? Calculate $P(M')$ as well.
- Assume R is normally distributed with $\mu = 107$ and $\sigma = 22$, calculate $P(90 \leq R \leq 120)$.
- Calculate $P(90 \leq R \leq 120)$ empirically using your data. How accurate were the assumptions about the distribution of R in 2b? Which of the assumptions do you consider the most suspect? Explain.
- Using your data, calculate $P(120 \leq R|M')$

step a

```
male_or_not <- ifelse(movies$director_gender == "male", 1, 0)
male_or_not[is.na(male_or_not)] <- 0
probability_male <- mean(male_or_not)
probability_male

## [1] 0.8477723

female_or_not <- ifelse(movies$director_gender == "female", 1, 0)
female_or_not[is.na(female_or_not)] <- 0
```

```
probability_female <- mean(female_or_not)
probability_female

## [1] 0.0990099
```

Your Answer:

The probability that a director is a male $P(M)=0.8477723$. The probability that a director is not a male (is a female) is $P(M')=0.0990099$.

step b

```
pnorm(120, mean= 107, sd = 22) - pnorm(90, mean = 107, sd = 22)

## [1] 0.5028674
```

Your Answer:

According to normal distribution, the probability that runtime, denoted by R , is between 90 and 120 minutes is 0.5028674.

step c

```
R <- movies$runtime
conditions_met_r <- R[ R >= 90 & R <= 120]
length(conditions_met_r) / length(R)

## [1] 0.6571782
```

Your Answer:

Empirically, the probability of the runtime being between 90 and 120 minutes lies higher than the probability in case of normal distribution, at $P(90 \leq R \leq 120) = 0.6571782$. Which probability is more trustworthy depends on what probability it is supposed to represent. If it is supposed to represent the probability within this sample, empirical probability is the most trustworthy. However, if it is supposed to represent the probability of the entire population, the probability according to normal distribution would be the most accurate, assuming that runtime is in fact distributed normally.

step d

```
#Probability of runtime more or equal to 120 minutes
meq_120_min <- R[R>=120]
prob_120_min <- length(meq_120_min)/length(R)

#Probability that a director is female
probability_female

## [1] 0.0990099

#Probability of intersection of female director and runtime
intersect_fem_r <- sum(movies$runtime>=120 & movies$director_gender == "femal
```

```
e", na.rm = TRUE)
prob_intersect <- length(intersect_fem_r)/length(movies$runtime)
intersect_fem_r

## [1] 15

#Probability of the union of female director and runtime >= 120 min
prob_union <- prob_120_min + probability_female - prob_intersect
prob_union

## [1] 0.2970297
```

Your Answer:

The probability of the union between a runtime of 120 minutes or more and a female director is 0.2970.

3 For this question, you will again use the profitability ratio PR. Make sure that PR is defined the same way as in question 3d of week 1. For this question, you will assume that your data set is the full population.

- What is the mean of the variable PR in your data set?
- Create a new data set, called `movies_sample`. Make sure that it is a random sample of your data set of 25 movies. What is the mean of PR in this random sample? How does it compare to the mean of PR in 3a?
- In a for loop, create 100 different samples of 25 movies, as in b, and estimate the mean within each sample. Save the mean of each sample in a vector called `sample_means`. So the first position of the vector would have the mean of the first sample, the second position the mean of the second sample, etc. Print this vector.
- Summarize and make a histogram of `sample_means`. What is the mean, standard deviation and shape of its distribution?
- Using the formulas of the *Central Limit Theorem*, give the distribution of the sample mean of PR in a sample of 25 movies from your population (including its expected value and standard deviation). How do these values compare to your answers at d.)?

step a

```
new_mean <- mean(movies$PR, na.rm = TRUE)
new_mean

## [1] 2.864599
```

Your Answer:

The mean of the variable PR is approximately 2.8646

step b

```
movies_sample <- sample_n(movies, 25)
mean(movies_sample$PR, na.rm = TRUE)
```

```
## [1] 3.812465
```

Your Answer:

The sample mean of variable PR from a sample of 25 random observations is approximately 2.3770. This is approximately 0.5 lower than the population mean.

step c

```
sample_means <- numeric(100)
for (i in 1:100) {
  current_sample <- sample_n(movies, 25)
  sample_means[i] <- mean(current_sample$PR, na.rm = TRUE)}
print(sample_means)

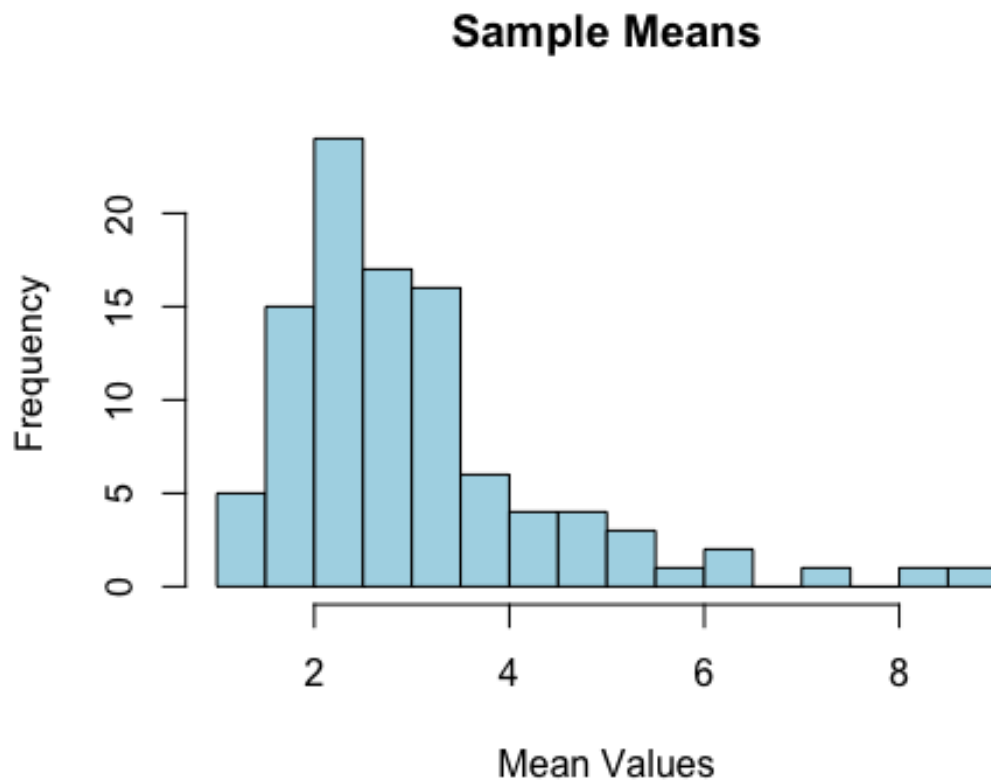
## [1] 2.262572 1.751018 2.202660 2.516784 1.764892 3.136770 4.929546 2.468
462
## [9] 5.191921 3.216551 2.954082 7.254199 2.614143 3.699928 4.766782 2.354
270
## [17] 4.817680 2.452567 2.051598 3.462158 3.096300 6.116623 2.349112 3.357
297
## [25] 2.901001 2.861922 2.273144 5.212466 3.510769 3.079164 6.302409 3.680
703
## [33] 3.090494 1.848437 1.730474 3.114650 4.237235 2.784097 2.410191 2.420
815
## [41] 1.870948 1.890777 1.957019 4.105775 2.345475 2.271837 2.698637 8.863
453
## [49] 3.206382 4.311074 2.747866 3.216705 2.416767 2.701518 2.532009 2.045
871
## [57] 2.392241 3.659708 2.131893 2.296803 2.221239 2.003645 2.647267 3.016
218
## [65] 3.085396 2.500617 1.019515 2.615069 2.978661 4.856941 2.632955 2.367
954
## [73] 1.680243 3.742253 1.805749 5.307583 2.683518 4.199175 3.212267 3.567
193
## [81] 1.472766 1.849842 1.655053 1.458542 2.174238 2.128337 1.700368 1.078
550
## [89] 8.283744 1.999244 1.335026 3.279673 2.440740 1.881419 1.810963 3.376
502
## [97] 3.278239 2.195680 5.738694 2.522116
```

Your Answer:

(no idea what to say)

step d

```
#Create Histogram
hist(sample_means, main = "Sample Means", xlab = "Mean Values", col = "lightb
lue", breaks = 16)
```



```
#Compute descriptive statistics
mean(sample_means)

## [1] 3.017118

sd(sample_means)

## [1] 1.412922

kurtosis(sample_means)

## [1] 7.020601

skewness(sample_means)

## [1] 1.827101
```

Your Answer:

For clarity, the histogram employs bins with a size of 0.5. The mean of the distribution of the sample means is 2.7636 and the standard deviation is 1.0590. Visually, the means concentrate between 1.5 and 3.5 with tails to 1 and to 5.5 and one “outlier” at 7.5-8. Mathematically, the kurtosis is 7.046 and the skewness is 1.6461. This means that the distribution has a heavy tail to the right and is asymmetric.

step e

```
PR_mean <- mean(movies$PR, na.rm = TRUE)
PR_sd <- sd(movies$PR, na.rm = TRUE)/sqrt(25)
PR_mean

## [1] 2.864599

PR_sd

## [1] 1.085395
```

Your Answer:

The population mean is 2.8646, while the mean of the sample means is 2.7636 and the population standard deviation for a sample of 25 observations is 1.0854, while it is 1.0590 according to the sample mean vector. These values are very similar, but slightly different because the sample size is too small. This is because the distribution of the sample means is asymmetric, causing the Central Limit Theorem to require at least 30 observations per sample to make a valid approximation of the population statistics.

Week 3

1. For the next part of the assignment, assume that the movies in your data frame are a random sample of a larger population of movies.
 - a. Create a new data set that only includes movies that are of the genre “Thriller”. For these thriller movies, give a 99 percent confidence interval for the variable *vote_average*.
 - b. According to IMDB, the average movie is rated with a 6.5. Using your data, test for the null hypothesis that thriller movies are at least as good as the average movie, against the alternative hypothesis that thriller movies are rated worse than the average movie. Test this hypothesis using an appropriate five-step procedure. What is your p-value and what do you conclude?

step a

```
thriller_movies <- filter(movies, movies$genre == "Thriller")
t.test(thriller_movies$vote_average, conf.level = 0.99)$conf.int

## [1] 5.675877 6.063778
## attr(,"conf.level")
## [1] 0.99
```

Your Answer:

The confidence interval of the population mean of vote average with a 99% confidence interval is (5.6759,6.064). This means that the population mean of the vote average is 99% certain to be between 5.6759 and 6.064.

step b

```
t.test(thriller_movies$vote_average, mu = 6.5, alternative = "less")

##
## One Sample t-test
##
## data:  thriller_movies$vote_average
## t = -8.5103, df = 115, p-value = 3.739e-14
## alternative hypothesis: true mean is less than 6.5
## 95 percent confidence interval:
##      -Inf 5.992615
## sample estimates:
## mean of x
##  5.869828
```

Your Answer:

Step 1: State the hypotheses: $H_0: \mu \geq 6.5$ and $H_a: \mu < 6.5$. μ here is the population mean of the vote average of thriller movies. Step 2: Specify the decision rule: this is a one-sided (left-tailed) test and since no significance level has been given, the default of $\alpha = 0.05$ will be used. So, the decision rule is that if $p\text{-value} < 0.05$ the null hypothesis will be rejected. Step 3: Collect data and calculate: this can easily be done with the `t.test()` function and the specification `alternative = "less"` to make this a left-tailed test. Step 4: the t-test gives a p-value of $3.739e-14$, which is a number extremely close to zero, this means that even with 99% certainty ($\alpha = 0.01$) the null hypothesis should be rejected, as the probability of observing the sample mean given that the null hypothesis is true is so small, that it is nearly impossible that this is accidental. Therefore, the null hypothesis should be rejected. Step 5: Take action: reject the null hypothesis. From the extremely low p-value we can conclude with almost full certainty that the average rating of a thriller movie is not equal or higher than that of the average movie, which is 6.5.

2

- How many movies in your sample had a female lead actor? How many had a male lead actor?
- Test the null hypothesis that for any given movie the chance that the lead actor is male or female is equal. Clearly state your null hypothesis and alternative hypothesis, and your chosen significance level. Use an appropriate R function to test your hypothesis, and state your conclusion. Also give the p-value that corresponds to your test statistic.
- What is the variance of the profitability ratio (PR) for movies with a male lead actor? What is the variance of profitability ratio for movies with a female lead actor? Make sure that PR is defined in the same way as in week 1.
- Assume that the variance of PR for movies with a male lead actor in your data is the same as the variance for movies with a male lead actor in your population (that is, there is no sampling uncertainty in this estimate). Now, set up a hypothesis test to

test for the null hypothesis that the PR for movies with a female lead is equal to the variance for movies with a male lead actor. Clearly state your null hypothesis and alternative hypothesis, and your chosen significance level. Calculate your test statistic and tell how your test statistic is distributed, and state your conclusion. Also give the p-value that corresponds to your test statistic.

step a

```
movies %>% count(first_actor_gender)

## # A tibble: 3 × 2
##   first_actor_gender     n
##   <chr>             <int>
## 1 female             216
## 2 male               558
## 3 <NA>               34
```

Your Answer:

In the sample, 216 movies have a female lead actor and 558 movies have a male lead actor. There are 34 movies for which the gender of the lead is marked as not available.

step b

```
#Remove NAs
gender_clean <- na.omit(movies$first_actor_gender)
length(gender_clean)

## [1] 774

#Compute prop.test (using the numbers computed in step a)
prop.test(x=216, n=774, p=0.50, conf.level=0.95)

##
## 1-sample proportions test with continuity correction
##
## data: 216 out of 774, null probability 0.5
## X-squared = 150.23, df = 1, p-value < 2.2e-16
## alternative hypothesis: true p is not equal to 0.5
## 95 percent confidence interval:
## 0.2480000 0.3123661
## sample estimates:
##           p
## 0.2790698
```

Your Answer:

In this assignment, the NAs are removed, reducing the number of observations in the sample from 808 to 774. We use the proportional distribution of male and female in the sample to assess the chance. $H_0: \pi = 0.5$ and $H_1: \pi \neq 0.5$, with significance level $\alpha = 0.05$. This is a two-tailed test, testing the probability that the proportion of females in the sample

is correct, given that the population proportion is 0.5. Given the chosen significance level, the p-value should be below 0.05 in order to reject the null hypothesis. The p-value is extremely close to zero ($p\text{-value} < 2.2e - 16$), meaning it is less than the significance level of 0.05. In conclusion, we reject the null hypothesis that the chance that the lead actor is male or female is equal for any movie with 95% certainty (although the p-value is so low that it could also be 99% certainty)

step c

```
#Variance of PR for female Lead movies
female_PR <- subset(movies, first_actor_gender == "female", select =PR)
female_var <- var(female_PR$PR, na.rm= TRUE)
female_var

## [1] 27.12278

#Variance of PR for male Lead movies
male_PR <- subset(movies, first_actor_gender == "male", select = PR)
male_var <- var(male_PR$PR, na.rm= TRUE)
male_var

## [1] 31.33558
```

Your Answer:

The sample variance of the profitability ratio of a movie with a female lead is 27.12278 and the sample variance of the profitability ratio of a movie with a male lead is 31.33558.

step d

```
#The needed variables
n <- length(female_PR$PR)
female_var

## [1] 27.12278

male_var

## [1] 31.33558

#The chi square value
chi_square <- (n-1)*female_var/male_var
chi_square

## [1] 186.0951

#Computing the p-value
p_value <- 2*min(pchisq(chi_square, df = n-1), 1 - pchisq(chi_square, df = n-1))
p_value

## [1] 0.152929
```

Your Answer:

The hypotheses are: $H_0: \sigma_f^2 = \sigma_m^2 = 31.33558$ and $H_1: \sigma_f^2 \neq \sigma_m^2 = 31.33558$, with a significance level of $\alpha = 0.05$ (95% certainty). The test statistic needed for test is a chi-square test statistic, this test statistic follows a asymmetric and positive distribution. The chi-square-value is $\chi^2 = 186.0951$ and the p-value is $p - value = 0.152929$ and since $0.152929 > 0.05$, the probability that the difference between the sample statistic and the population statistic is due to luck is too big to reject the null hypothesis, so we do not reject the null hypothesis. Therefore, we reject the alternative hypothesis that the variance of PR for female leads is not equal to the variance of PR for male leads and we do not reject the null hypothesis that they are equal to each other, because the p-value is too high.

3

- Create a new variable called genre2. Make sure that the three most popular genres remain the same, but that all the other genres get categorized into one genre called "Other". Report a frequency table of this new variable genre2.
- Create a scatter plot with each type of genre2 on the x axis and the mean of PR within that genre on the y-axis. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. Which movie genre has the highest PR? Which has the lowest?
- Recreate the scatter plot, but now add bars around each point, indicating the 95% confidence interval.
- Now, create a new data frame which is a random subsample of $n=100$ of your full data frame. Recreate the plot in c.) How do the confidence intervals compare to the confidence intervals you plotted in c.)?

step a

```
#Find three most popular genres
movies %>% count(genre, sort = TRUE)

## # A tibble: 14 × 2
##   genre      n
##   <chr>    <int>
## 1 Drama    402
## 2 Comedy   186
## 3 Thriller  116
## 4 Action    41
## 5 Adventure  16
## 6 Family    11
## 7 Horror    11
## 8 Documentary  9
## 9 Fiction    5
## 10 Music     4
## 11 <NA>       3
## 12 Crime     2
```

```
## 13 Romance      1
## 14 Western      1

#Create new variable
movies <- movies %>% mutate(genre2 = case_when(
  genre %in% "Drama" ~ "Drama",
  genre %in% "Comedy" ~ "Comedy",
  genre %in% "Thriller" ~ "Thriller",
  TRUE ~ "Other"))
#Frequency table
table(movies$genre2)

##
##   Comedy   Drama   Other Thriller
##    186     402     104     116
```

Your Answer:

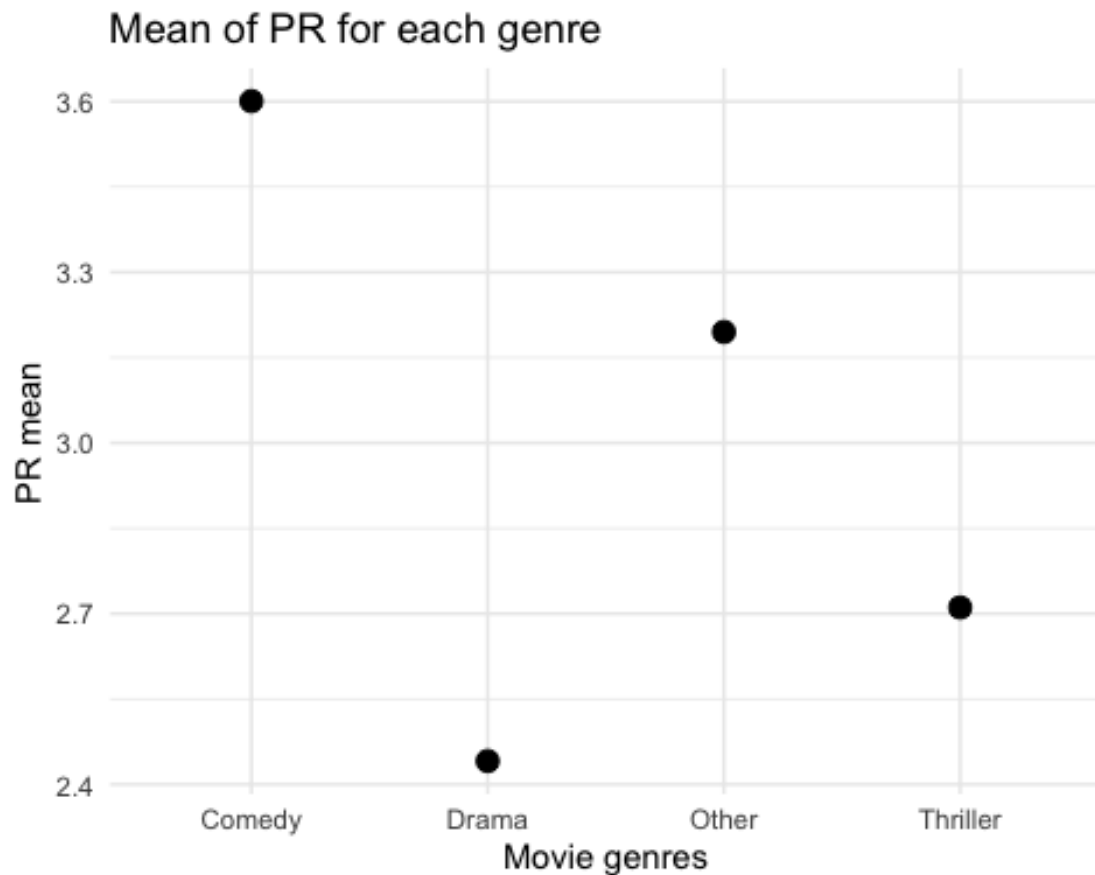
The three most popular genres are Drama, Comedy, and Thriller.

step b

```
#Get the mean of PR for each genre
mean_genre <- movies %>% group_by(genre2) %>% summarise(mean_PR2 = mean(PR, n
a.rm = TRUE))
mean(mean_genre$mean_PR2)

## [1] 2.986888

#Make scatterplot
ggplot(mean_genre, aes(x=genre2, y=mean_PR2)) +
  geom_point(size = 3) +
  labs(title = "Mean of PR for each genre", x = "Movie genres", y = "PR mean")
+
  theme_minimal()
```

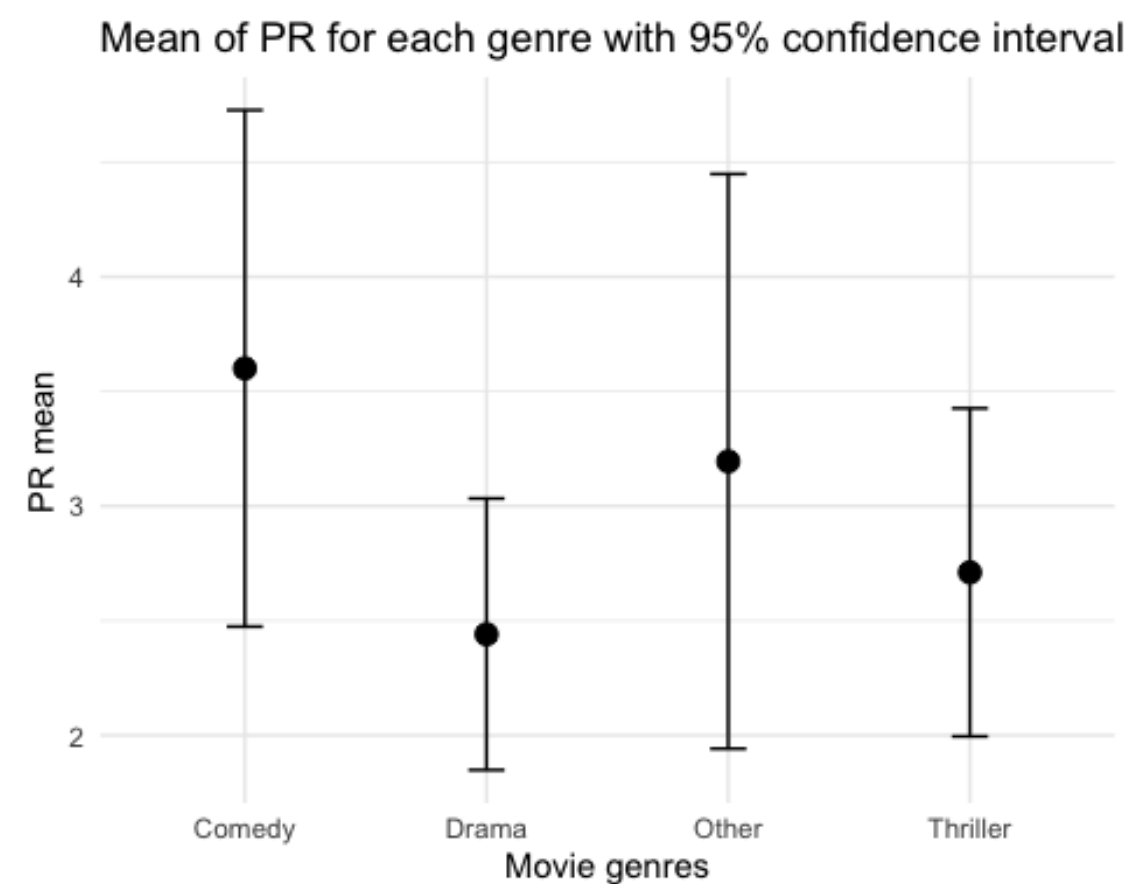


Your Answer:

Comedy has the highest PR of all the genres, with a PR of approximately 3.60 and Drama has the lowest PR of all the genres, with a PR of approximately 2.44.

step c

```
#Remake dataset mean_genre with confidence intervals
mean_genre_ci <- movies %>% group_by(genre2) %>%
  summarise(
    n = sum(!is.na(PR)),
    mean_PR2 = mean(PR, na.rm = TRUE),
    sd_PR = sd(PR, na.rm = TRUE),
    se_PR = sd_PR/sqrt(n),
    lower_ci = mean_PR2 - qt(0.975, df = n-1)*se_PR,
    upper_ci = mean_PR2 + qt(0.975, df = n-1)*se_PR
  )
#Make scatterplot with updated information
ggplot(mean_genre_ci, aes(x = genre2, y = mean_PR2))+
  geom_point(size = 3) +
  geom_errorbar(aes(ymin = lower_ci, ymax = upper_ci), width = 0.15) +
  labs(title = "Mean of PR for each genre with 95% confidence interval", x =
"Movie genres", y = "PR mean") + theme_minimal()
```

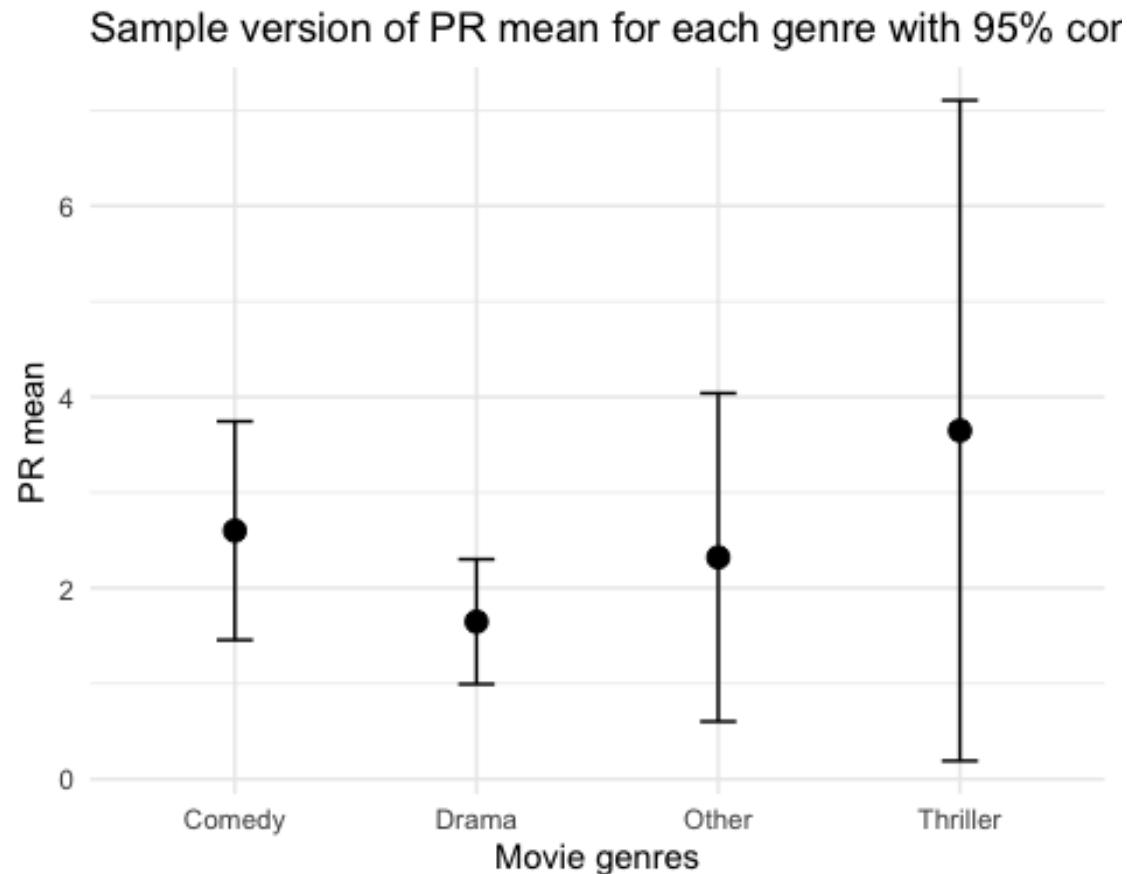


Your Answer:

step d

```
#Make sample of movies with n=100
sample_movies <- sample_n(movies, 100)
#Generate dataset with necessary values for scatterplot
sample_movies <- sample_movies %>% group_by(genre2) %>%
  summarise(
    n = sum(!is.na(PR)),
    mean_sample_PR = mean(PR, na.rm = TRUE),
    sd_sample_PR = sd(PR, na.rm = TRUE),
    se_sample_PR = sd_sample_PR / sqrt(n),
    low_ci_sample = mean_sample_PR - qt(0.975, df = n-1)*se_sample_PR,
    up_ci_sample = mean_sample_PR + qt(0.975, df = n-1)*se_sample_PR
  )
#Plot the generated variables in a scatterplot
ggplot(sample_movies, aes(x= genre2, y= mean_sample_PR )) +
  geom_point(size = 3) +
  geom_errorbar(aes(ymin = low_ci_sample, ymax = up_ci_sample), width = 0.15)
+
  labs(title = "Sample version of PR mean for each genre with 95% confidence")
```

```
interval", x = "Movie genres", y = "PR mean") +  
  theme_minimal()
```



Your Answer:

Both the confidence intervals and the means of the PR for each genre from the sample are very different from those of the whole dataset. Note that the y-axis of this scatterplot is distributed slightly different from the one at 3c, so visual comparison needs to be done carefully. The confidence interval of the PR mean of Thriller has increased drastically and that of Drama as well, but not as much as Thriller's. The confidence interval for Other decreased and that of Comedy stayed approximately the same. The PR mean of Thriller increased, but for all other genres the means decreased.